PRODUCT REQUIREMENTS · v2.0

Facility & Estate Management Platform

A 2-in-1, AI-powered platform for managing facilities, estates and communities — predicting problems before they happen and automating the busywork. **Version 2.0 — developer-ready draft.**

Version	2.0 — Developer-ready draft
Prepared by	Unrealabs (Choplife Technologies Ltd)
Prepared for	7 Fifteen Capital
Audience	Product, engineering & QA teams; 7 Fifteen Capital stakeholders
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Status	Living document — refined during discovery; material changes via Change Request
Related agreement	Design & Development Services Agreement (commercial terms map to Schedule A)
Classification	Confidential

1. Executive summary

7 Fifteen Capital requires a modern platform to run facilities, estates and residential communities end-to-end. Unrealabs proposes a 2-in-1, AI-powered platform that unifies facility operations and estate/community management on a single shared core — predicting problems before they happen, automating the busywork, and giving managers, residents, owners and investors one modern experience across mobile, web and admin.

The product ships as a **multi-tenant platform** with two functional modules — **Facility Management** and **Estate Management** — on a shared, AI-enabled core covering payments, multi-platform apps, document intelligence and an investor off-plan dashboard. Delivery is a one-time build priced at **USD 300,000** over an indicative **≈24–26 weeks (July – December 2026)**.

What changed in v2.0: this revision makes the PRD buildable. Every requirement now carries a priority (MoSCoW), a user story and testable acceptance criteria; the domain model, tenancy model, permissions matrix and key workflow states are defined; NFRs are quantified; AI features specify inputs/outputs, human-in-the-loop behaviour, fallbacks and proposed benchmarks; and an open-questions register assigns the decisions still needed before or during discovery. Items marked *(proposed)* or *(TBC)* are baselines for estimation and build — they are confirmed with 7 Fifteen Capital at discovery and do not silently become contract terms.

2. Product vision

> *“One intelligent platform that runs facilities and estates end-to-end — predicting problems before they happen, automating the busywork, and giving managers, residents and owners a single, modern experience.”*

- **Two worlds, one product** — unifies facility operations and estate/community management across mobile, web and admin.
- **AI from day one** — built in, not bolted on, to cut reactive maintenance and lift collections and satisfaction.
- **Real-time visibility** — clear, live insight for owners and investors across the whole portfolio.

3. Problem & opportunity

Management today is fragmented and largely manual — spreadsheets, WhatsApp groups, paper logs and disconnected tools. The predictable results:

- **Reactive, not preventive** — assets fail before anyone acts, driving up downtime and cost.
- **Slow request handling** — issues get lost; the right technician or vendor is hard to assign.
- **Leaky billing** — service charges go unpaid and reconciliation is painful.
- **Poor experience** — little transparency, slow communication, no self-service.
- **No single truth** — data is scattered, so reporting and decisions are guesswork.

The opportunity: one intelligent platform running both facility operations and estate/community life on a shared core — problems predicted rather than reacted to, routine work automated, and every stakeholder working from the same live data.

4. Glossary

Term	Definition
Tenant / Organisation	A client organisation onboarded onto the platform (e.g. an estate operator or facility owner managed by 7 Fifteen Capital). All data is scoped to a tenant.
Estate	A residential/mixed community managed under a tenant: units, residents, amenities, security.
Facility	A managed building or site with assets, maintenance and compliance obligations.
Unit / Space	An addressable space (apartment, office, zone) within an estate or facility.
Work order (WO)	A tracked maintenance job — reactive (from a request) or planned (from a PPM schedule).
PPM	Planned preventive maintenance — recurring, scheduled maintenance plans.
Service charge	A recurring or one-off charge billed to a unit/resident.
Arrears	Unpaid charges past their due date.
Off-plan	Property sold before/during construction; tracked on the investor dashboard.
RAG	Retrieval-augmented generation — AI answers grounded in the tenant's own documents.
HITL	Human-in-the-loop — an AI suggestion a person reviews, approves or overrides.
NDPA	Nigeria Data Protection Act — governing privacy, consent and biometric/CCTV features.

5. Target users

User	Module	What they need
Facility Manager	Facility	Oversee assets, maintenance, vendors and compliance from one dashboard.
Maintenance Technician	Facility	Receive, prioritise and resolve work orders; log inspections on mobile.
Property Owner / Executive	Facility	Portfolio-level visibility, reports and ROI.
Estate Manager	Estate	Run estate operations: billing, communication, bookings, security.
Resident	Estate	Pay charges, raise requests, book facilities, get updates, use the marketplace.
Worker / Security	Estate	Access control, SOS response, patrols and directory.
Vendor / Contractor	Both	Receive jobs, submit updates/invoices, list services on the marketplace.
Investor	Shared	Track off-plan sales, occupancy and returns via a dedicated dashboard.

6. Product scope & prioritisation

6.1 In scope — one product, two modules, a shared core

Component	Coverage
Module 01 — Facility Management	Assets, maintenance, inspections, vendors, space and reporting.
Module 02 — Estate Management	Billing, residents, requests, bookings, security, marketplace.
Shared AI-enabled core	Multi-tenant deployment · payments · mobile + web + admin · AI copilot · document intelligence · investor dashboard.

6.2 Priority scheme (MoSCoW)

All Must/Should/Could items below are inside the contracted scope and price (§18); priorities set the **build order and MVP cut**, not what is bought. **Must** = MVP; the platform does not go live without it. **Should** = committed, follows MVP within the timeline. **Could** = built last; first to reflow if the timeline is squeezed (any descope via Change Request). **Won't (this phase)** = §6.3.

6.3 Out of scope (Won't — this phase)

- Features outside the agreed scope (added via Change Request, Clause 10).
- Bespoke AI model training beyond the foundation-model + RAG approach.
- Maintenance and support beyond the 30-day warranty (available under a separate retainer).
- Content creation and third-party API-driven changes.
- Hardware procurement (CCTV, IoT devices) — software integration only.

7. Roles & permissions matrix

Server-side, deny-by-default enforcement on every API (SC-01). Legend: **F** full control · **M** manage (create/edit within own tenant/estate) · **V** view · **O** own records only · **—** no access. Column key: Plat = Platform Admin (7 Fifteen Capital HQ), Exec = Owner/Executive, EstM = Estate Manager, FacM = Facility Manager, Tech = Technician, Res = Resident, Sec = Security, Ven = Vendor, Inv = Investor.

Capability	Plat	Exec	EstM	FacM	Tech	Res	Sec	Ven	Inv
Tenant onboarding & config	F	—	V	—	—	—	—	—	—
Users & roles	F	V	M	M	—	O	—	O	O
Assets & PPM plans	V	V	V	M	V	—	—	—	—
Work orders	V	V	M	M	O	O	—	O	—
Inspections & compliance	V	V	V	M	O	—	—	—	—
Charges & invoicing	V	V	M	—	—	O	—	—	—
Payments (pay in-app)	—	—	—	—	—	O	—	—	—
Announcements	V	V	M	M	—	V	V	—	—
Bookings & amenities	V	V	M	—	—	O	V	—	—
Security & SOS	V	V	M	—	—	O	M	—	—
Marketplace	V	V	M	—	—	O	—	O	—
Investor projects	F	V	—	—	—	—	—	—	O
Reports & analytics	F	V	M	M	—	—	—	—	V

Fine-grained permissions per endpoint are finalised at design; this matrix is the authoritative baseline.

8. Domain model & multi-tenancy

8.1 Core entities

Entity	Purpose	Key relationships
Tenant	Client organisation; the isolation boundary for all data.	Has estates/facilities, users, configuration, payment account.
Estate / Facility	A managed community or building.	Belongs to tenant; has buildings, units, amenities, staff.
Unit / Space	Addressable space in the hierarchy estate → building → floor → unit/zone.	Links residents, assets, charges, bookings.
User + Role(s)	Any actor; roles per §7, scoped to tenant (and estate where relevant).	Resident links user → unit(s); vendor links user → vendor org.
Asset	Equipment/infrastructure under management.	Belongs to unit/space; has PPM plans, work orders, documents.
PPM plan	Recurring maintenance definition (frequency, checklist, lead time).	Generates work orders for an asset/category.
Work order	A maintenance job with lifecycle per §10.	Links asset/unit, requester, assignee (tech or vendor), costs, evidence.
Inspection	Checklist run with pass/fail items and evidence.	Failed items raise work orders; feeds compliance register.
Vendor	External contractor organisation.	Receives work orders; has documents, invoices, ratings, listings.
Charge / Invoice	Billable item per unit/resident with due date.	Has payments (possibly partial); drives arrears aging.
Payment	Gateway transaction reconciled to invoices/orders.	Webhook-confirmed; idempotent; per-tenant settlement account.
Announcement	Broadcast or targeted message.	Delivered via notifications engine; tracks delivery/read.
Booking	Reservation of an amenity.	Links resident, amenity, optional payment; approval per amenity rules.
Visitor pass	Time-bound access code/QR for a guest.	Issued by resident; verified by security; logged.
SOS event	Panic alert with acknowledge/resolve trail.	Links resident/unit, responders, timestamps.
Document	Stored file with permissions.	Indexed by document intelligence (AI-S-03).
Listing / Order	Marketplace product/service and purchase.	Vendor listing → resident order → payment → fulfilment.
Investor project / Lead	Off-plan project with sales pipeline.	Drives investor dashboard and AI projections.
AI interaction log	Every AI request/response, confidence, override.	Per tenant; powers audit, tuning and cost metering.

8.2 Tenancy model & isolation requirements

- **Hierarchy:** Platform → Tenant (organisation) → Estates/Facilities → Buildings → Units → Residents/Assets. Every domain record carries its tenant scope.
- **Hard isolation:** no query, file, search index, AI retrieval or report may cross tenants below Platform Admin. Verified by automated cross-tenant access tests in CI (release-blocking).
- **Isolation strategy:** shared database with tenant-scoped rows enforced at application and database layer (e.g. row-level security), tenant-scoped object storage and per-tenant vector-store namespaces *(proposed — confirm as Architecture Decision ADR-01, open question Q2)*.
- **Tenant onboarding:** Platform Admin creates the organisation, then configures estates/units (bulk import), branding, payment settlement account, languages and enabled features.
- **Per-tenant configuration:** branding, notification templates, amenity rules, charge structures, AI feature flags and usage caps.
- **Cost metering:** AI, SMS and email usage metered per tenant (assumption §15: usage costs are billed to the Client).

9. Functional requirements

Each requirement carries an ID, a MoSCoW priority, a primary user story and key acceptance criteria (AC). ACs are the definition of done for build and QA; they are proposed baselines, confirmed per feature during discovery/design. Requirements marked *(new in v2.0)* were implicit in v1.0 and are now explicit — they add no scope beyond what the v1.0 features already required to function.

9.1 Module 01 — Facility Management

ID · Priority	Requirement & user story	Key acceptance criteria
FM-01 Must	**Asset management & register** Story — As a Facility Manager, I maintain a register of every asset so identity, location and history are trackable.	• Asset CRUD: name, category, serial/tag, location (estate→building→unit/zone), status, purchase & warranty dates, photos, documents. • Assets link to work orders and inspections; service history viewable per asset. • Archive (no hard delete); all changes audit-logged (SC-11). • Bulk import via CSV template.
FM-02 Must	**Preventive maintenance & scheduling** Story — As a Facility Manager, I define recurring PPM plans so work is scheduled before failure.	• PPM plans: frequency, checklist, assignee rules, lead time, per asset or category. • Scheduler auto-generates work orders ahead of due date; calendar view; overdue flagged. • Completion requires checklist + evidence photo (configurable).
FM-03 Must *(new in v2.0)*	**Work-order management** Story — As a Technician, I receive, update and close work orders on mobile so nothing gets lost.	• Lifecycle states and transitions exactly per §10 with per-priority SLA timers. • Assign to internal technician or vendor; reassignment and every state change tracked and notified. • Capture photos, notes, parts and costs; requester notified on state changes. • Offline updates supported per §12.1.
FM-04 Should	**Space management** Story — As a Facility Manager, I manage the space hierarchy so occupancy and allocation are always current.	• Estate→building→floor→unit/zone hierarchy maintained once, shared with the Estate module. • Occupancy/allocation status per space; spaces link to assets, bookings and residents.
FM-05 Must	**Vendor management** Story — As a Facility Manager, I onboard vendors and route jobs to them so external work is controlled.	• Vendor profiles: services, coverage, documents (insurance/certifications) with expiry alerts. • Vendors receive assignments and submit updates & invoices via the vendor portal. • Performance metrics: jobs completed, SLA hit rate, rating.
FM-06 Should	**Inspection & compliance tracking** Story — As a Facility Manager, I schedule inspections so compliance is evidenced, not assumed.	• Inspection templates with checklist items (pass/fail/severity); scheduled and ad-hoc runs on mobile with photo capture. • Failed items auto-raise work orders (FM-03). • Compliance register shows status and expiry per obligation.
FM-07 Must	**Reporting & analytics** Story — As an Executive, I see portfolio performance so decisions stop being guesswork.	• Dashboards: open/overdue WOs, PPM compliance %, downtime, cost by category/estate. • Filters by estate/date; export CSV/PDF; role-scoped visibility per §7.
FM-08 Must	**Full mobile access** Story — As a Technician, I do my whole job from my phone, including in poor-connectivity areas.	• All technician and FM day-to-day flows usable on Android/iOS. • Push notifications for assignments and escalations; camera-first capture.

9.2 Module 02 — Estate Management

ID · Priority	Requirement & user story	Key acceptance criteria
EM-01 Must	**Service charge & billing** Story — As an Estate Manager, I issue charges and track collection so cash flow is predictable.	• Charge setup per unit type/size: recurring cycles, one-off levies, penalties/discounts (configurable). • Invoices generated and delivered in-app + email/SMS; residents pay in-app via gateway (SC-02). • Payments auto-reconcile to invoices; partial payments, receipts and arrears aging supported. • Statements per unit/resident; accounting export.
EM-02 Must	**Residents & visitor management** Story — As an Estate Manager, I onboard residents and control guest access so I know who is on the estate.	• Resident onboarding (invite + profile) linked to unit(s); move-in/move-out; household members supported. • Visitor invites generate time-bound codes/QR verified by Security at the gate; visitor log retained per NDPA retention policy (§12).
EM-03 Must	**Maintenance request portal** Story — As a Resident, I raise a request with photos and track it to resolution without chasing anyone.	• Request: category, description, photos → ticket ID with visible status. • Requests feed the shared work-order pipeline (FM-03) with estate routing. • Resident notified on each state change and can rate the resolution.
EM-04 Must	**Announcements & communication** Story — As an Estate Manager, I reach the right residents on the channels they actually read.	• Broadcast and targeted (estate/building/audience) sends via in-app, push, SMS, email per user preference (SC-09). • Delivery/read statistics visible to managers. • AI drafting/translation (AI-E-04) always requires manager approval before send.
EM-05 Should	**Facility booking** Story — As a Resident, I book shared amenities without a WhatsApp back-and-forth.	• Amenity setup: capacity, hours, rules, optional fees. • Availability calendar; auto- or manager-approval per amenity; fees paid via gateway; cancellation per policy.
EM-06 Must	**Security & SOS alerts** Story — As a Resident, I trigger an SOS and know security has seen it.	• SOS alerts security desk and designated responders with unit/location; acknowledge → responding → resolved with timestamps (§10). • Patrol and incident logging for security staff; all security events audit-logged. • SOS submission works on poor connectivity with SMS fallback *(proposed)*.
EM-07 Should	**Vendor / staff directory** Story — As a Resident, I find approved vendors and estate staff without asking around.	• Searchable directory with services and contact routes; visibility rules per role (§7).
EM-08 Should	**Document management** Story — As an Estate Manager, I publish estate documents once and the right people can always find them.	• Upload/organise with folder and permission control; residents see documents shared to their audience. • All stored documents indexed for AI document intelligence (AI-S-03).
EM-09 Should	**Marketplace** Story — As a Resident, I buy services from approved vendors inside the app.	• Vendors list products/services; residents browse, order and pay via gateway. • Order lifecycle (placed → accepted → fulfilled → closed) with notifications. • Commission/payout model per commercial decision — open question Q3 blocks build of payouts.

9.3 Shared core platform

ID · Priority	Requirement & user story	Key acceptance criteria
SC-01 Must	**Multi-user profiles & RBAC** Story — As any user, I see and do only what my role allows — enforced by the server, not the UI.	• Roles and access per §7, tenant- and estate-scoped; deny-by-default on every API. • Session and password policy; MFA for admin roles *(proposed)*.
SC-02 Must	**Payments integration** Story — As a Resident, I pay any charge in-app and it reconciles itself.	• Gateway abstraction (provider per §13) supporting cards, transfers and USSD where available. • Webhook-confirmed payments with idempotency and retries; refund/void; per-tenant reconciliation report.
SC-03 Must	**Core integrations** Story — As the platform, I send SMS/email and render maps through swappable providers.	• SMS/email behind an abstraction with template management and delivery-failure handling/logs; maps for locations.

ID · Priority	Requirement & user story	Key acceptance criteria
SC-04 Should	**E-commerce / marketplace infrastructure** Story — As the platform, I provide shared catalog/order/payment services that EM-09 consumes.	• Catalog, cart/order and payment services shared between marketplace and paid bookings.
SC-05 Must	**Mobile apps (Android, iOS)** Story — As a user, I get the same product on Android and iOS.	• Single cross-platform codebase (framework decision at kickoff); Android AAB + iOS builds. • Distribution: Play Store + TestFlight/IPA per v1.0 — production App Store release is open question Q11. • Push via FCM/APNs; crash and performance reporting wired in.
SC-06 Must	**Web app & admin dashboard** Story — As an operator, I run everything from a browser.	• Responsive web app for residents/managers; platform admin console for tenant management, configuration and audit views.
SC-07 Must	**Multi-tenant deployment** Story — As Platform Admin, I onboard a new organisation without touching code, and tenants can never see each other.	• All §8.2 isolation requirements met; cross-tenant access tests pass in CI (release-blocking). • Tenant onboarding, configuration and per-tenant usage metering implemented.
SC-08 Should	**Investor off-plan dashboard** Story — As an Investor, I track my projects' sales, occupancy and returns.	• Off-plan projects with sales pipeline, occupancy and returns; investors see only their projects. • AI projections (AI-S-04) always labelled as estimates with assumptions shown.
SC-09 Must *(new in v2.0)*	**Notifications engine** Story — As a user, I get the right alerts on the right channel — and can control them.	• Unified service for push/SMS/email/in-app used by all modules (single event bus). • Per-user channel preferences and quiet hours; per-tenant templates; delivery logging.
SC-10 Must *(new in v2.0)*	**Audit logging** Story — As a compliance owner, I can reconstruct who did what, when.	• Immutable log of auth events, permission changes, financial actions, security events and AI overrides — actor, tenant, timestamp. • 12-month retention *(proposed)*; queryable from the admin console.
SC-11 Should	**Landing page** Story — As a prospect or investor, I learn about the platform and register interest.	• Public marketing page with lead capture into the investor pipeline *(proposed)*.

10. Key workflows & states

State machines below are the contract between backend, mobile and QA. Only the listed transitions are legal; every transition is timestamped, actor-attributed and notified via SC-09. SLA values are proposed defaults, configurable per tenant.

Workflow	States	Rules & SLAs (proposed)
Work order	New → Triaged → Assigned → In Progress → On Hold → Resolved → Verified → Closed; branches: Rejected, Cancelled	AI triage suggests category/priority/assignee; human confirms (AI-F/E triage). SLA timers per priority (e.g. Emergency: respond 2h / resolve 24h; Routine: 72h). Resolved → Verified by requester or manager; auto-close after 5 days *(proposed)*.
Service charge	Draft → Issued → Partially paid → Paid; Issued → Overdue → In arrears → Escalated	Reminder schedule (e.g. –3d, due, +7d, +21d) via SC-09; AI timing suggestions (AI-E-03) require manager approval. Escalation policy per tenant.
Booking	Requested → Approved / Auto-approved → Confirmed → Completed; branches: Declined, Cancelled	Auto-approval per amenity rules; payment (if fees) required to confirm; cancellation window per policy.
SOS event	Raised → Acknowledged → Responding → Resolved → Reviewed	Acknowledge target < 60s during manned hours *(proposed)*; unacknowledged alerts escalate to backup responders; full timeline retained for review.
Marketplace order	Placed → Accepted → Fulfilled → Closed; branches: Declined, Refunded	Vendor must accept within SLA or order auto-expires; refunds via gateway; payout model pending Q3.

11. AI requirements

AI is the differentiator, delivered on a foundation-model + RAG architecture. Every AI feature must define the four columns below before build: what goes in, what comes out, where the human sits, and what happens when the AI is unavailable. Benchmarks are proposed acceptance baselines (per §15: acceptance is against agreed benchmarks, not 100% accuracy) — final numbers are agreed at discovery and measured on an eval set built with 7 Fifteen Capital.

11.1 Facility Management AI

ID · Priority	Capability: inputs → outputs	Human-in-the-loop & fallback	Proposed benchmark (TBC)
AI-F-01 Should	**Predictive maintenance** — asset attributes, WO/failure history, sensor feeds where available → ranked failure-risk list + suggested PPM dates as draft schedule entries.	FM approves before any schedule changes. Runs in shadow mode until data volume is sufficient. Fallback: standard PPM calendar unaffected.	After 3–6 months of data: precision ≥ 70% on 30-day failure window.
AI-F-02 Should	**Smart work-order triage** — request text/photos/category → suggested category, priority and best-fit assignee/vendor with confidence.	Dispatcher confirms or overrides; overrides logged for tuning. Fallback: manual triage queue.	Top-1 category accuracy ≥ 85%; assignee suggestion acceptance ≥ 70%.
AI-F-03 Should	**Vision-based inspections** — inspection photos → detected defects with severity + auto-drafted report.	Inspector edits and approves every report. Fallback: manual checklist flow.	Defect recall ≥ 80% on agreed defect classes (pilot set).
AI-F-04 Should	**Analytics & anomaly alerts** — cost/energy/usage series → plain-language anomaly cards with drill-down.	Confirm/dismiss feedback loop tunes sensitivity. Fallback: static threshold alerts.	False-positive rate ≤ 30% after tuning period.
AI-F-05 Should	**Facility-manager copilot** — natural-language queries over tenant-scoped data → answers with sources; drafted actions (e.g. raise WO).	Every drafted action requires explicit user confirmation. Fallback: normal navigation and search.	Grounded-answer accuracy ≥ 90% on eval set.

11.2 Estate Management AI

ID · Priority	Capability: inputs → outputs	Human-in-the-loop & fallback	Proposed benchmark (TBC)
AI-E-01 Should	**Resident support chatbot** — resident question + estate RAG corpus → grounded answer with source, or escalation.	No source → must refuse and escalate to a ticket (EM-03). Corpus curated by the Estate Manager. Fallback: FAQ + ticket.	Routine-query deflection ≥ 60%; grounded-answer accuracy ≥ 90%.
AI-E-02 Should	**Smart request triage** — photo-based classification and auto-routing of resident requests (shared service with AI-F-02).	Same HITL and fallback as AI-F-02.	Shared benchmark with AI-F-02.
AI-E-03 Should	**Billing & arrears intelligence** — billing/payment history → anomaly flags, defaulter-risk scores, reminder-timing suggestions.	Managers approve reminder campaigns; no automated penalties ever. Fallback: fixed reminder schedule.	Risk model AUC ≥ 0.75 after two billing cycles.
AI-E-04 Should	**Communication assistant** — drafts announcements and translates into configured languages.	Manager approval required before any send. Fallback: manual compose.	Human review pass rate ≥ 90%.
AI-E-05 Could	**Marketplace intelligence** — listing auto-generation, personalised recommendations, fraud flags.	Fraud flags go to a human review queue — never auto-block. Fallback: manual listings, no recommendations.	Fraud-flag precision ≥ 60% at review *(pilot)*.
AI-E-06 Could *(optional)*	**AI security & access** — CCTV anomaly detection and face/plate recognition.	Prerequisites: NDPA compliance, explicit consent and DPIA sign-off (Q5). Builds only after sign-off; integrates with client hardware (hardware out of scope).	Defined during DPIA; not in default acceptance.

11.3 Shared platform AI

ID · Priority	Capability: inputs → outputs	Human-in-the-loop & fallback	Proposed benchmark (TBC)
AI-S-01 Should	**Platform-wide copilot** — one conversational assistant across both modules and admin (extends AI-F-05).	Action confirmation as AI-F-05. Fallback: standard UI.	Shared benchmark with AI-F-05.
AI-S-02 Should	**NL search & reporting** — questions across data → answers and reports in plain language with the underlying records viewable.	Every figure traceable to source records. Fallback: standard reports (FM-07).	Query success rate ≥ 80% on agreed test set.
AI-S-03 Should	**Document intelligence** — OCR + field extraction + semantic search across document management.	Extracted fields editable before save. Fallback: manual entry and keyword search.	Field extraction ≥ 95% printed / ≥ 85% scanned documents.
AI-S-04 Should	**Investor-dashboard AI** — sales/occupancy forecasting, lead scoring, ROI projections.	All outputs labelled as estimates with assumptions shown; not presented as financial advice.	Forecast error (MAPE) reviewed quarterly vs actuals.
AI-S-05 Could	**Recommendation & fraud engine** — shared personalisation and anomaly service behind AI-E-05 and bookings.	As AI-E-05.	As AI-E-05.
AI-S-06 Must	**AI architecture & guardrails** — LLM + RAG vector store, prompt/cost controls, logging; the platform every AI feature runs on.	See §11.4 — release-blocking requirements.	Zero cross-tenant retrieval in isolation tests (hard requirement).

11.4 AI guardrails & design requirements (release-blocking)

- **Tenant isolation:** retrieval, embeddings and context are hard-scoped to the tenant; automated cross-tenant leakage tests run in CI.
- **Provider abstraction:** LLM/OCR/translation behind an abstraction with health checks; on provider failure, features degrade to their listed fallback — core flows never block on AI.
- **Cost controls:** per-tenant and per-feature usage caps with alerts and kill-switch per feature (usage billed to Client per §15).
- **Prompt-injection defence:** user- and document-supplied content is treated as data; the assistant cannot be instructed by it to exceed the user's permissions (§7).
- **Logging & privacy:** all AI interactions logged (AI interaction log, §8.1) with PII minimised/redacted in logs.
- **UX states:** every AI surface shows loading, confidence, sources and error states; every suggestion is review/approve/override — AI is assistive, not autonomous.

12. Non-functional requirements (quantified)

Targets marked *(proposed)* are engineering baselines to design and test against; confirm with 7 Fifteen Capital at discovery.

Area	Requirement	Measurable target (proposed)
Performance	Fast load on mobile networks; smooth under concurrent load.	API P95 < 500 ms; key screens interactive < 3 s on 4G and usable on 3G.
Reliability	High availability; AI fails safely.	99.5% monthly availability for core flows; AI outage never blocks core; backups with RPO ≤ 24 h, RTO ≤ 4 h.
Security	RBAC, encryption, certified gateway, audit logs, tenant isolation.	OWASP ASVS L2; TLS 1.2+ in transit, AES-256 at rest; MFA for admin roles; audit log per SC-10; cross-tenant tests release-blocking.
Privacy & compliance	NDPA compliance; consent for biometric/CCTV.	Consent records + processing register; DPIA before any AI-E-06 work; per-entity retention schedule; data residency per Q7.

Area	Requirement	Measurable target (proposed)
Scalability	Grows with tenants, units and users.	Launch sizing to confirm (Q8) — design target: 20 tenants / 10,000 units / 50,000 users / 1,000 concurrent; documented horizontal scale path.
Accessibility	Mobile-first, responsive, connectivity-tolerant.	WCAG 2.1 AA for web; offline behaviour per §12.1.
Localization	Multilingual for a diverse user base.	English at launch; additional languages per Q6; all UI strings externalised; AI translation for communications (AI-E-04).
Maintainability	Modular, shared design system, documented handover.	Modular services; CI/CD; unit-test coverage ≥ 70% on core domains; runbooks + handover docs per §16.

12.1 Offline & poor-connectivity behaviour

- Read caching for dashboards, work-order lists and key resident screens.
- Queued writes for work-order updates, inspection logs and SOS (SMS fallback for SOS, *proposed*), with visible sync status.
- Conflict resolution is server-authoritative; overwritten offline edits are surfaced to the user and audit-logged.

13. Platform, integrations & environments

13.1 Footprint

- **Mobile** — Android (APK/AAB) and iOS (TestFlight/IPA); store release plan per Q11.
- **Web** — product web app plus admin dashboard.
- **Environments** — dev, staging and production with CI/CD promotion; staging seeded with realistic multi-tenant test data.

13.2 Integration decisions (Client selects at kickoff; third-party fees per §15)

Area	Need	Candidate providers (illustrative)	Decide by
Payments	Cards, transfers, USSD; webhooks; per-tenant settlement.	Paystack · Flutterwave	Kickoff (blocks EM-01)
SMS	Transactional + OTP delivery in Nigeria.	Termii · Twilio	Kickoff
Email	Transactional email at volume.	AWS SES · SendGrid · Mailgun	Kickoff
Maps	Location display and geocoding.	Google Maps Platform	Design phase
LLM / AI	Copilots, chatbot, triage, translation via provider abstraction.	Provider-agnostic (e.g. Anthropic, OpenAI) behind AI-S-06	Design phase
OCR / vision	Document extraction; inspection photo analysis.	Cloud vision API or equivalent	Design phase
Push	Mobile push notifications.	FCM · APNs	Design phase
Hosting	Cloud hosting; residency per Q7.	To be agreed with Client	Kickoff

14. Success metrics (KPIs) with traceability

Proposed measures, baselined and confirmed during discovery. Each KPI names the requirements that move it, so instrumentation is built in — every metric below must be reportable from the product itself (FM-07 / AI-S-02).

Category	Metrics	Driven by
Maintenance	% preventive vs reactive work · mean time to resolve · asset downtime reduction.	FM-02, FM-03, AI-F-01/02

Category	Metrics	Driven by
Estate finance	Service-charge collection rate · on-time payment % · arrears reduction.	EM-01, SC-02, AI-E-03
Service quality	Request resolution time · resident satisfaction score.	EM-03, FM-03, AI-E-02
Engagement	Resident app adoption · chatbot deflection rate · marketplace transactions.	EM-02/04/09, AI-E-01
AI performance	Triage accuracy · prediction precision · % queries resolved by AI.	§11 benchmarks, AI-S-06 logs
Commercial	Investor-dashboard usage · off-plan sales and occupancy.	SC-08, AI-S-04, SC-11

15. Assumptions & dependencies

15.1 General

- Development proceeds on approved designs and agreed requirements.
- Third-party fees (SMS, email, hosting, maps, payment gateway) are paid by the Client.
- App Store / Play Store publishing requires Client-owned developer accounts.

15.2 AI-specific

- **Usage costs** — LLM, vision, OCR and translation usage is billed to the Client (metered per tenant, §8.2).
- **Data dependency** — AI quality improves as data accumulates; data-hungry features (AI-F-01, AI-E-03) launch in shadow/tuning mode.
- **Assistive by design** — AI produces suggestions with human review; it is not a replacement for judgement.
- **Privacy & consent** — biometric/CCTV features are optional and NDPA-contingent (Q5).
- **Scope boundary** — foundation models + RAG; bespoke model training is separate.
- **Acceptance** — AI features are accepted against the agreed benchmarks of §11, not 100% accuracy.

16. Delivery plan & phase mapping

Indicative 6-month plan, **July – December 2026**; detailed roadmap confirmed at kickoff. Phases map requirements to the v1.0 milestones so the team always knows what “core development” means.

Phase	Weeks (indic.)	Focus	Requirement IDs
1 — Foundations & MVP	1–12	Discovery, brand, UX/UI; shared core; Must-priority features; AI guardrails scaffolding.	SC-01/02/03/05/06/07/09/10 · FM-01/02/03/05/07/08 · EM-01/02/03/04/06 · AI-S-06
2 — Committed scope	10–20	Should-priority features behind flags; AI feature set on the guardrails platform.	FM-04/06 · EM-05/07/08/09 · SC-04/08/11 · AI-F-01…05 · AI-E-01…04 · AI-S-01…04
3 — Complete & harden	18–26	Could items; performance/security hardening; UAT; deployment, go-live, integration & handover.	AI-E-05/06 (if Q5 signed off) · AI-S-05 · NFR verification · docs & runbooks

Process: Discovery → design (up to 3 revision rounds) → build → QA → client review (5 business days; deemed acceptance if no response) → deployment — with a **30-day post-deployment warranty** on reproducible bugs. Milestones: (1) discovery, brand & UX/UI kickoff · (2) core development · (3) testing & QA · (4) deployment · (5) go-live · (6) third-party integration & handover.

17. Open questions & decisions register

These block or shape build work. Each needs an owner and an answer by the phase shown; unresolved items default to the *(proposed)* baseline in this document.

#	Question	Why it matters	Owner	Needed by
Q1	Who exactly is a “tenant” — estates operated by 7 Fifteen Capital, or third-party operators too?	Shapes onboarding, branding and commercial model.	Client	Kickoff
Q2	Confirm data-isolation strategy (ADR-01: shared DB + row-level security vs schema-per-tenant).	Foundational architecture; expensive to change later.	Unrealabs	Kickoff
Q3	Marketplace commercial model: commission %, escrow, vendor payout rails and KYC.	Payouts have regulatory and gateway implications; blocks EM-09 payout build.	Client	Phase 2 start
Q4	SOS operating model: who responds, escalation chain, liability and manned hours.	Safety-critical flow; drives EM-06 SLAs and staffing.	Client	Phase 1
Q5	Proceed with AI security & access (AI-E-06)? Requires consent framework + DPIA.	NDPA-contingent; long lead time.	Client + Unrealabs	Phase 2
Q6	Language list for localization at launch.	Sizes translation and QA effort.	Client	Design phase
Q7	Hosting provider and data-residency requirement (Nigeria vs regional cloud).	NDPA posture, latency and cost.	Client	Kickoff
Q8	Launch sizing: tenants, units, users, expected concurrency.	Validates §12 scalability targets and load tests.	Client	Design phase
Q9	Any legacy data to migrate (spreadsheets, resident lists, asset registers)?	Migration is unscoped; may need a Change Request.	Client	Kickoff
Q10	Payment reconciliation source of truth (gateway settlement vs bank statements).	Defines EM-01 reconciliation rules.	Client	Phase 1
Q11	iOS distribution: v1.0 lists TestFlight/IPA — is a production App Store release in scope?	Affects review lead times and release plan.	Client + Unrealabs	Phase 1
Q12	Per-tenant AI/SMS budget caps and alert thresholds.	Client pays usage; caps prevent surprises.	Client	Phase 2

18. Commercials

Unchanged from v1.0. One package, one-time build fee; all figures in USD · multi-tenant · covers design, development, QA and delivery. Commercial terms map to Schedule A of the Design & Development Services Agreement.

What's included	Complete + AI
Brand identity (A1)	✓
Shared core — mobile, web, admin, payments	✓
Facility Management module	✓
Estate Management module	✓
AI capabilities	✓
QA, PM, deployment & 30-day warranty	✓
Indicative timeline	≈24–26 weeks
Total (one-time, USD)	**USD 300,000**

Component	Coverage	Price (USD)
Foundation	Brand (A1) + shared core + base delivery	70,000

Component	Coverage	Price (USD)
Facility module	Asset, maintenance, inspection, vendor, reporting	40,000
Estate module	Billing, residents, requests, bookings, security, marketplace	40,000
AI capabilities	Predictive maintenance, copilots, chatbot, doc intelligence, fraud	150,000
Total — Complete + AI	Everything, AI built in	**300,000**

- One-time fee in USD; excludes third-party running costs — hosting, SMS, email, maps, payment gateway and AI usage are billed separately to the Client.
- Up to 3 design revision rounds per module; 30-day warranty.
- Upgrades and scope changes via Change Request (Clause 10). Estimate valid for 30 days.

19. Approval & sign-off

This PRD is a living document and may be refined during discovery. Material changes to scope, features or assumptions are handled via Change Request under the Design & Development Services Agreement.

FOR 7 FIFTEEN CAPITAL (CLIENT)	
Name	
Title	
Signature	
Date	

FOR UNREALABS (CONTRACTOR)	
Name	**Samuel Omosehin**
Title	**Founder**
Signature	
Date	

Note: ongoing third-party running costs (hosting, SMS, email, maps, payment gateway and AI usage) are billed separately to the Client.

20. Change log

Version	Date	Change
1.0	Jul 2026	Initial PRD derived from the submission deck (prepared for Greenbox Ltd).
1.1	Jul 2026	Client corrected to 7 Fifteen Capital; deployment model changed to multi-tenant.
2.0	Jul 2026	Business-analysis revision for developer readiness: MoSCoW priorities, user stories and acceptance criteria on every requirement; glossary; roles & permissions matrix; domain model and tenancy/isolation requirements; new explicit requirements FM-03 work orders, SC-09 notifications, SC-10 audit logging; workflow state machines; AI specs with HITL, fallbacks and proposed benchmarks; quantified NFRs and offline behaviour; integration decision table; KPI traceability; phase-mapped delivery plan; open-questions register.